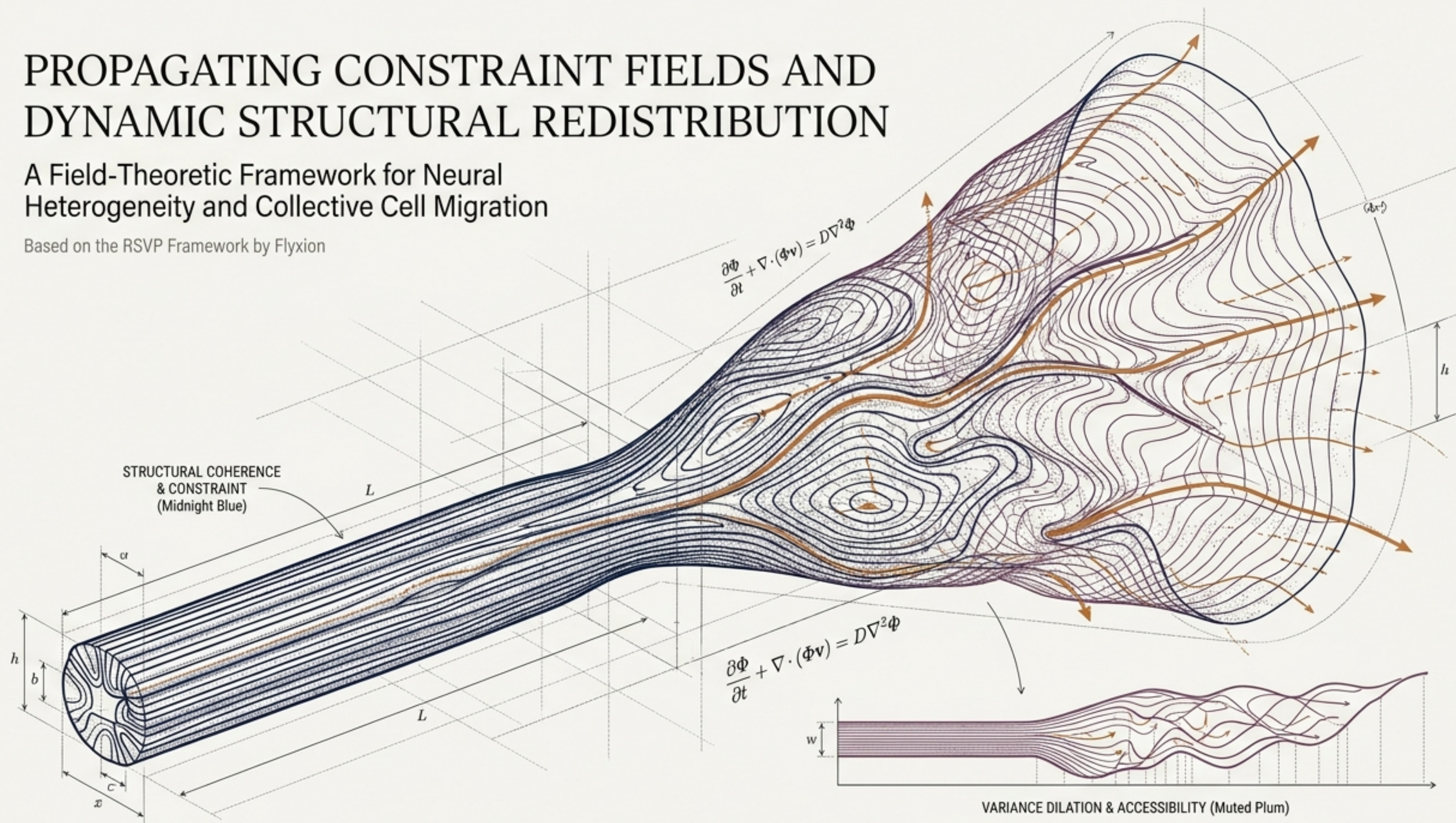


PROPAGATING CONSTRAINT FIELDS AND DYNAMIC STRUCTURAL REDISTRIBUTION

A Field-Theoretic Framework for Neural Heterogeneity and Collective Cell Migration

Based on the RSVP Framework by Flyxion



The Systemic Failure of Localist Models

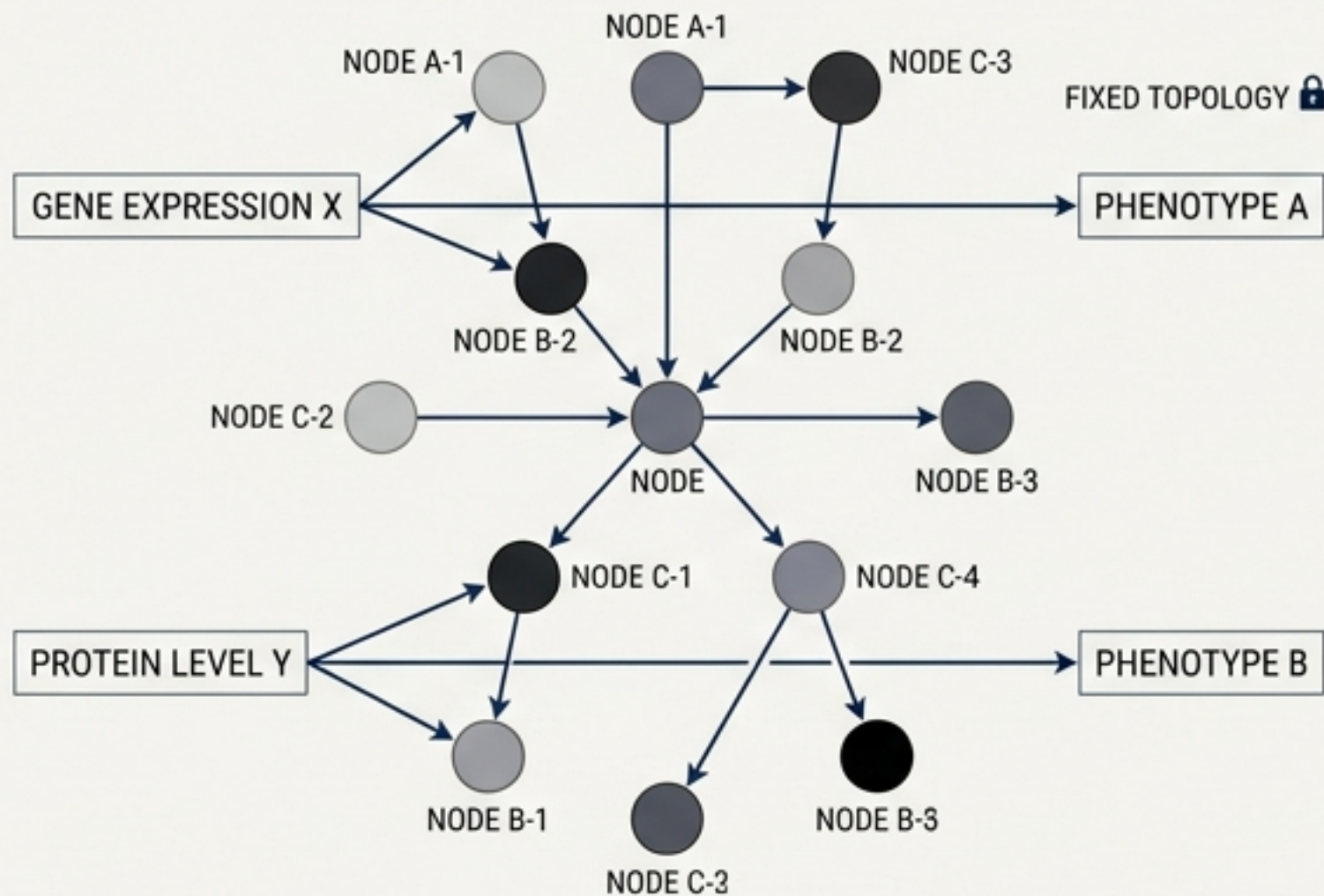
Standard entity-centered biological models break down when macroscopic behavior emerges from distributed interactions.

Entity-Centered Framework

Assumes stable categorical shifts.

Pathology = displaced mean.

Cells = independent agents.

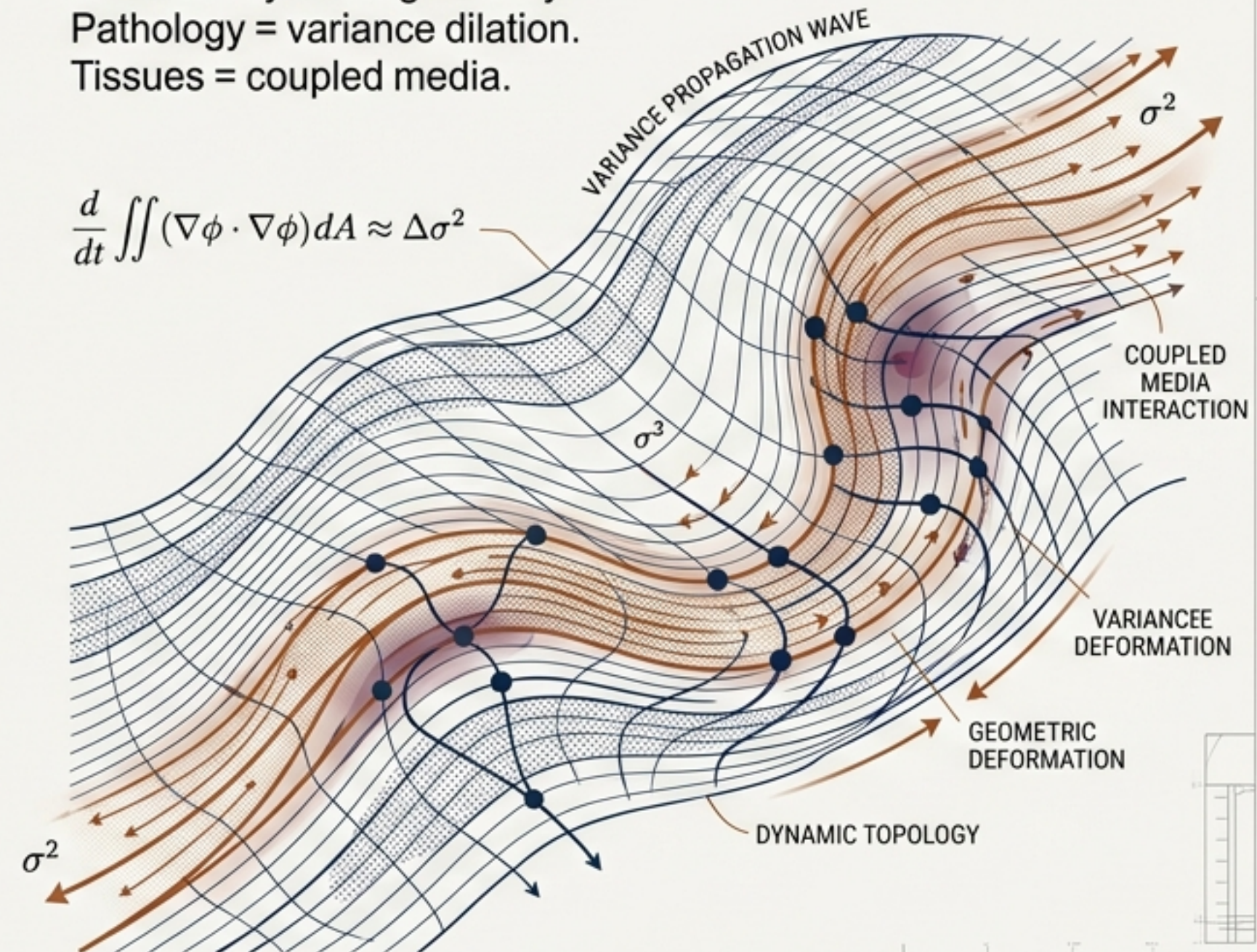


Constraint-Propagating Dynamics

Assumes dynamic geometry.

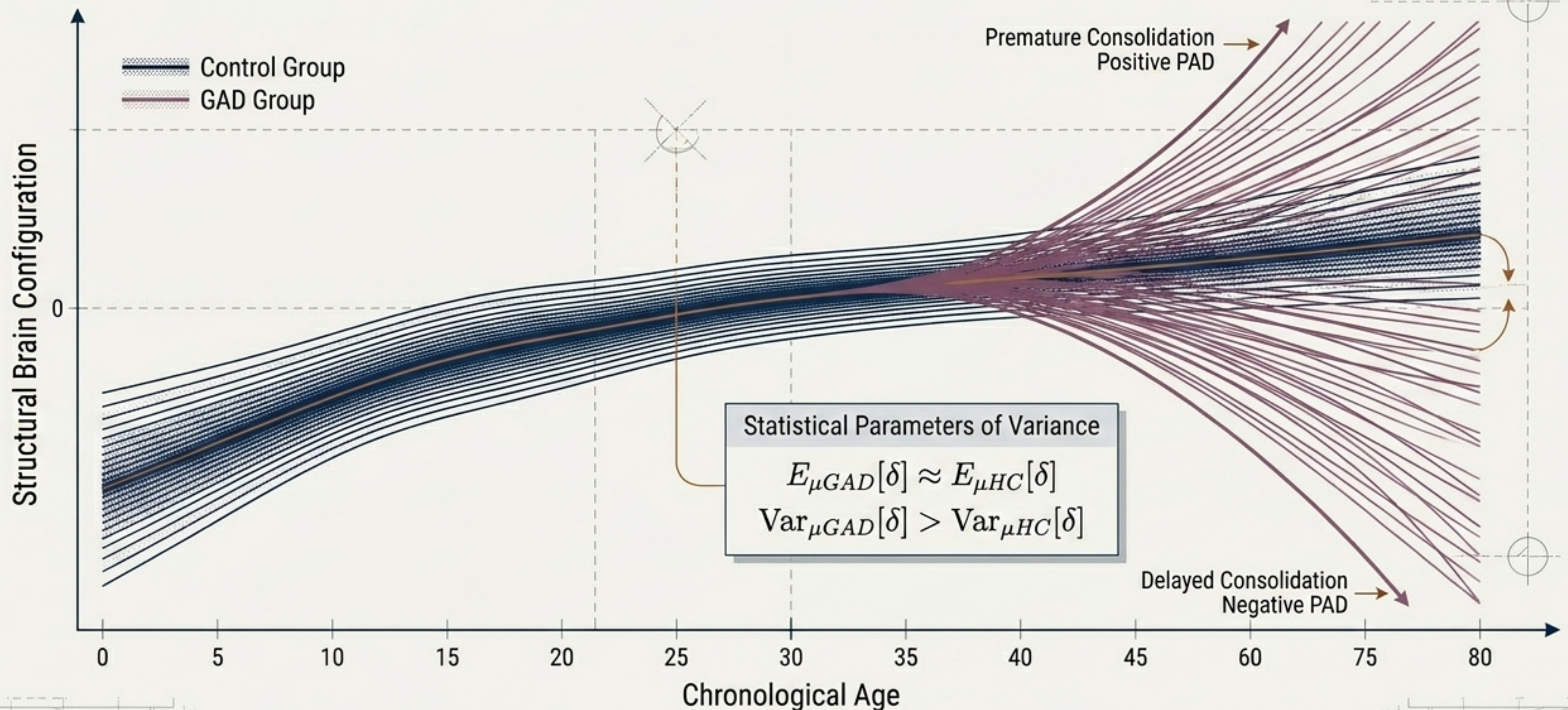
Pathology = variance dilation.

Tissues = coupled media.



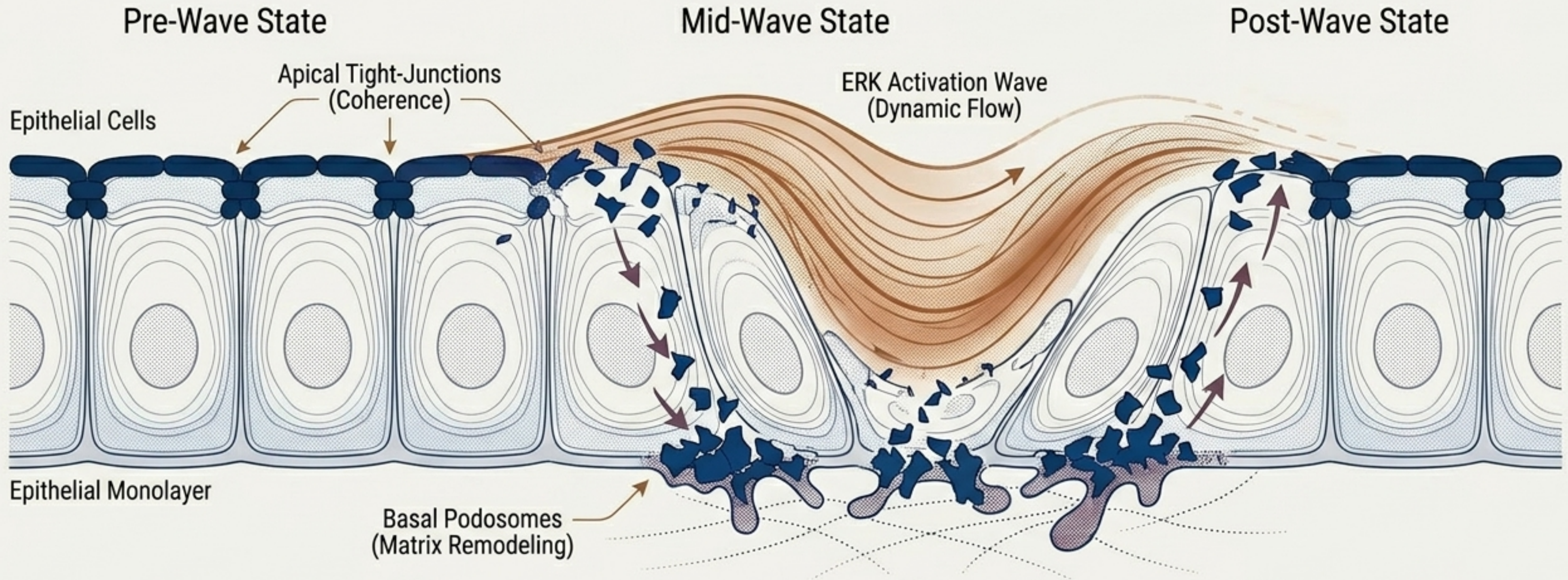
Anomaly I: The Geometry of Anxiety

CNN models of brain aging in GAD contradict the accelerated-aging hypothesis. The disorder expands the variance of structural states rather than shifting the mean.



Anomaly II: The Collective Tissue

During collective cell migration, the tight-junction protein ZO-1 is not a static barrier. It dynamically shuttles to basal podosomes in response to ERK activation waves.



The tissue is a coupled dynamical medium undergoing transient structural phase transitions, not independent cellular decisions.

The Structural Parallel

Despite vastly different scales, both anomalies reveal that localist models fail to capture distributed field-like dynamics.

Anomaly I (GAD Neural Trajectories)

Failed Assumption:

Disorder is a single point in developmental space.



Observed Reality:

Increased variance in Predicted Age Difference.



Distributed Dynamic:

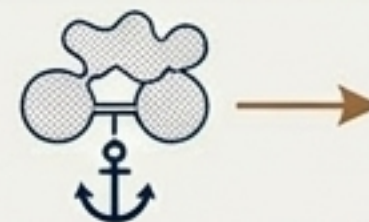
Trajectory-space dilation.



Anomaly II (ZO-1 & ERK Waves)

Failed Assumption:

Protein molecules possess stable functional properties.



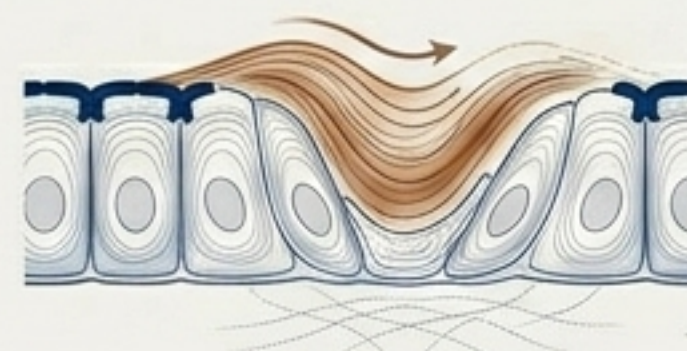
Observed Reality:

ZO-1 transiently reallocates constraint density.



Distributed Dynamic:

Traveling constraint-propagating fronts.

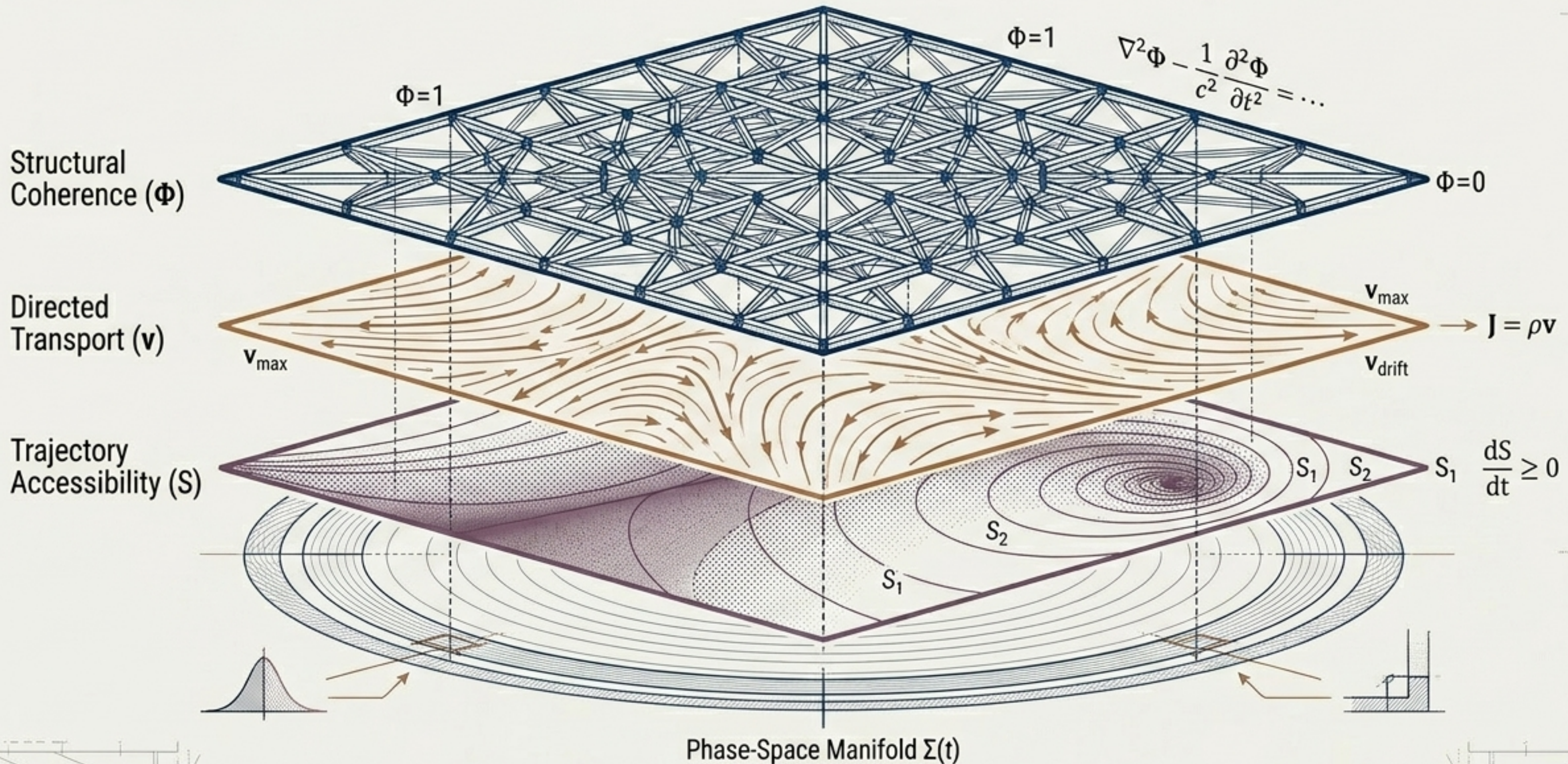


$$CR(X, t) = \int V(X') \cdot W(X - X') dt + \nabla \cdot (D \cdot \nabla C)$$

Both systems require an entirely new mathematical structure of Constraint Redistribution.

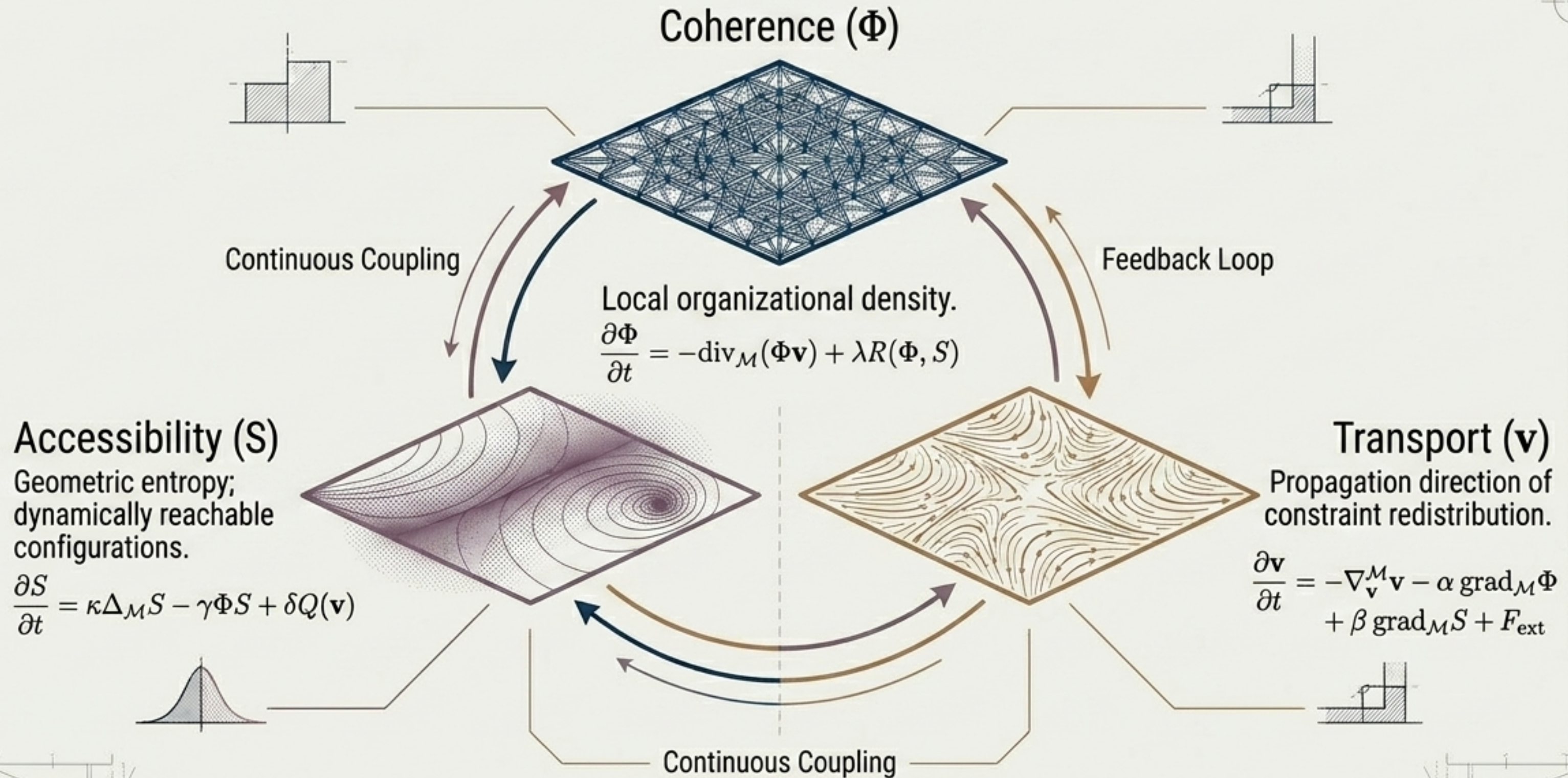
The Unifying Theory: Relativistic Scalar-Vector Plenum (RSVP)

Biology is dynamic admissibility geometry. The question is not which states a system can occupy, but which trajectory structures it can sustain.



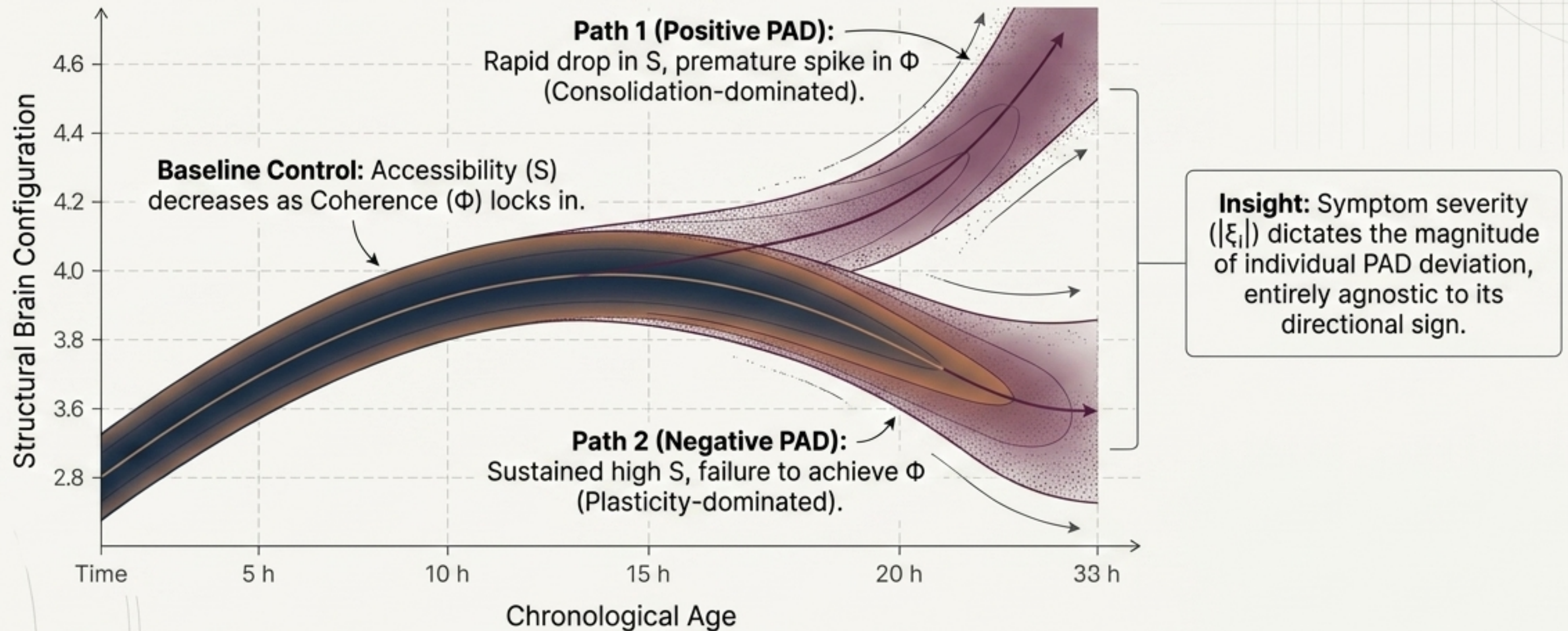
The Field Triple: $(\Phi, \mathbf{v}, S)$

The organizational state of the biological manifold evolves through three continuously coupled dynamical quantities.



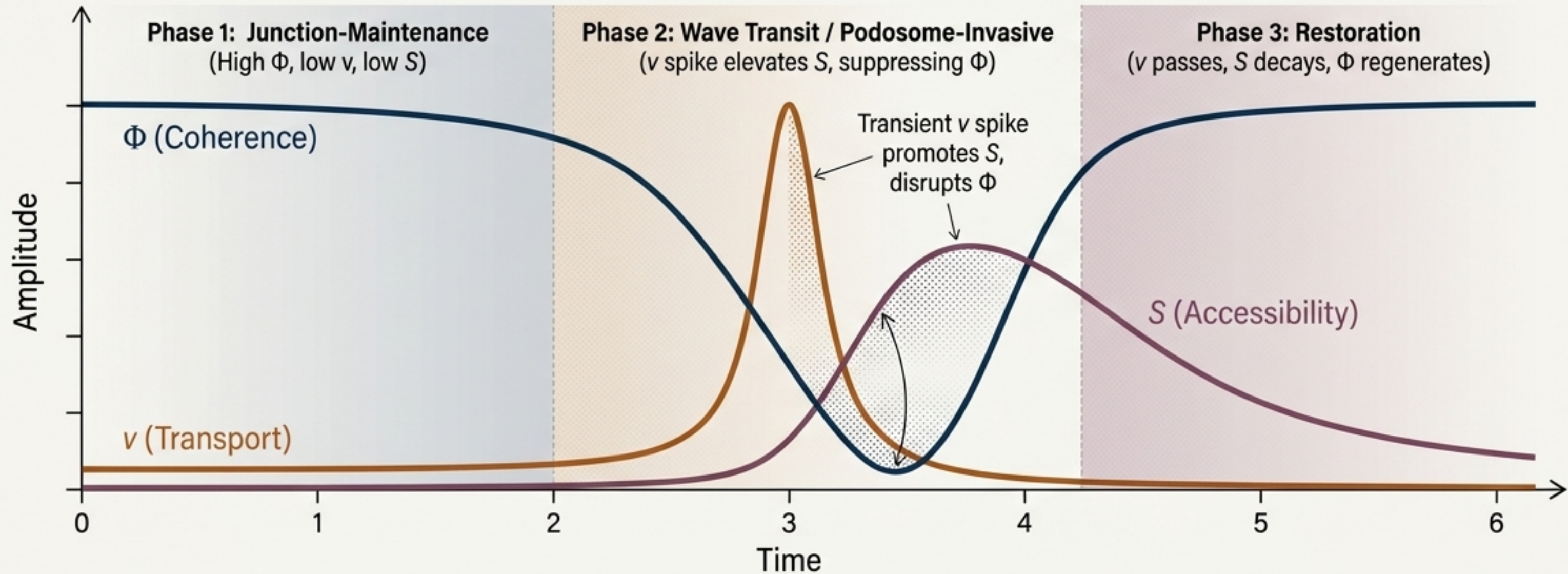
RSVP Application I: Neural Trajectories in GAD

Mapping the field triple onto developmental space explains variance amplification without shifting the mean.



RSVP Application II: Tissue Migration Dynamics

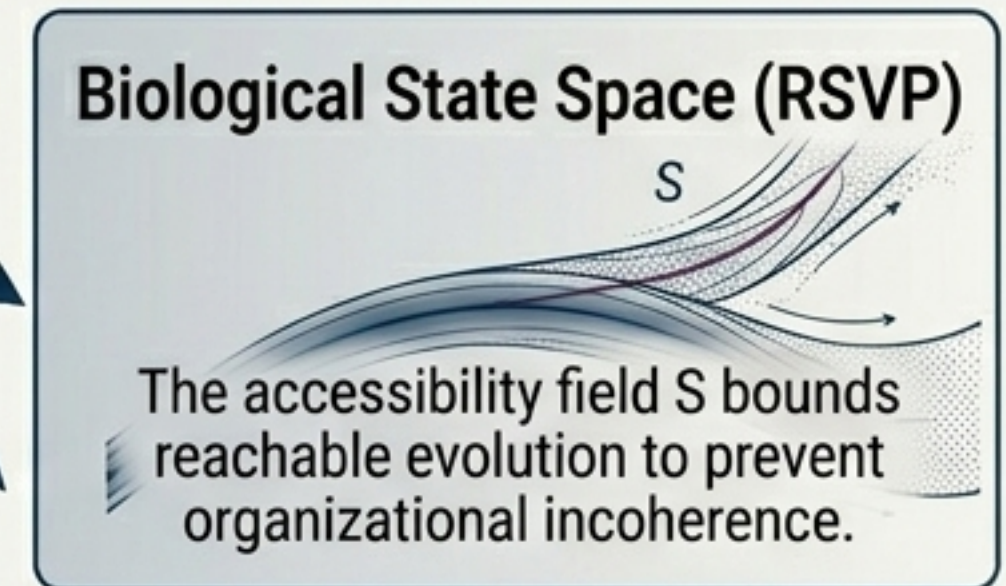
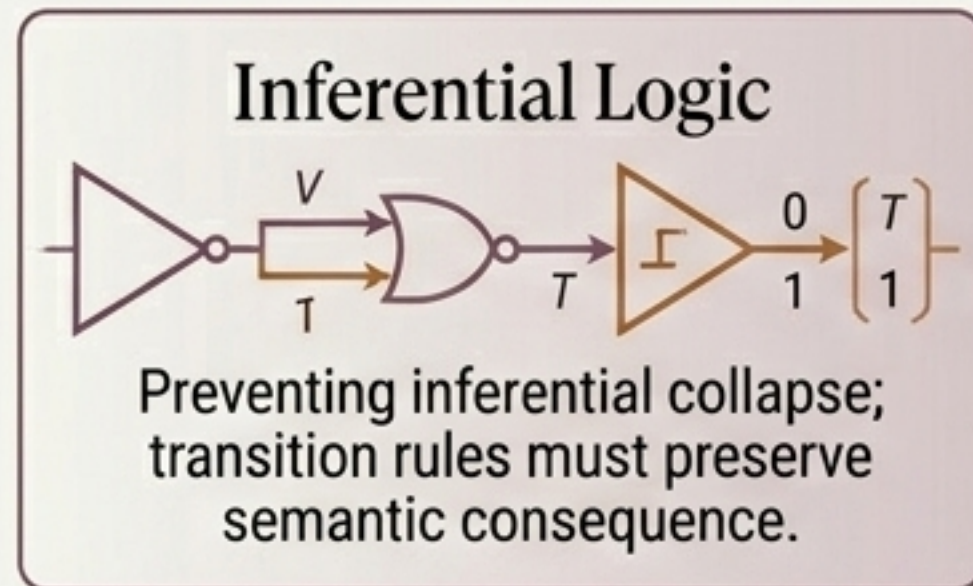
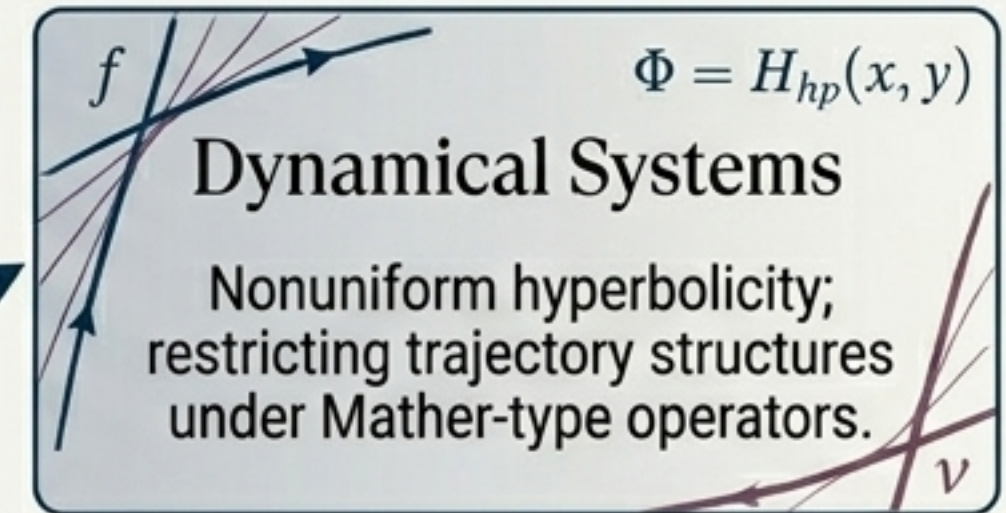
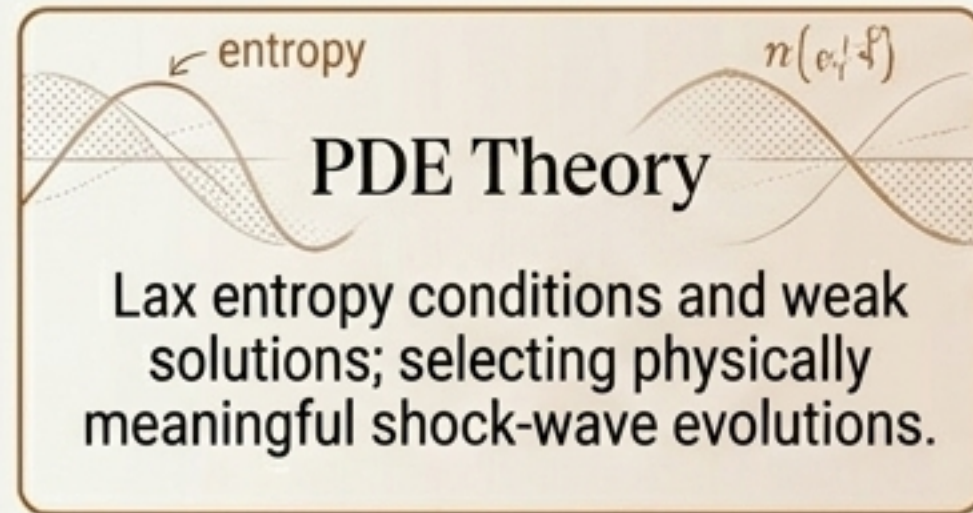
Wave-coupled phase transitions map perfectly to transient reallocation of constraint densities.



Key Concept: Migration requires transient, reversible constraint reallocation. Irreversible transitions cause tissue disintegration.

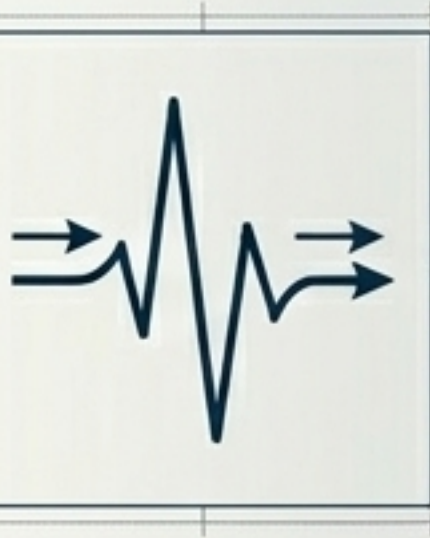
Admissibility as a Cross-Domain Invariant

Restricting formally possible trajectories to structurally coherent ones spans diverse mathematical domains.



The 4 Cross-Scale Structural Invariants

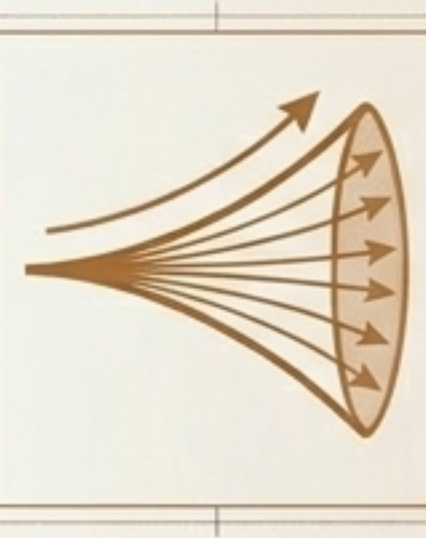
Universal rules governing distributed biological coordination,
independent of molecular substrate.



Constraint Propagation

Over Signal Diffusion

Complex behaviors are enabled by traveling boundaries between dynamical regimes, not mean-field intensity gradients.



Variance Amplification

Over Mean Displacement

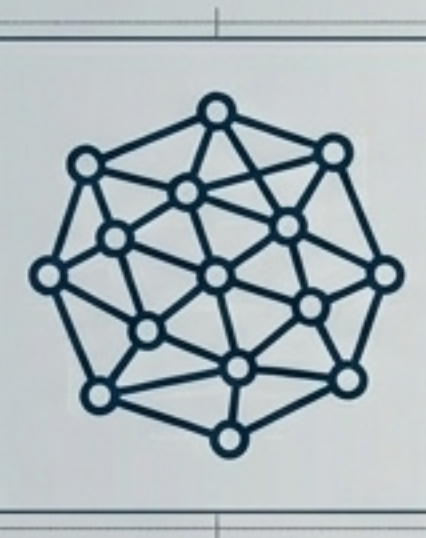
Altered organizational dynamics manifest as a widened distribution of accessible behaviors, not a shifted average.



Reversibility

As Organizational Competence

Health is defined by the capacity for controlled excursion and return (phase transitions), not rigid structural permanence.



Distributed Coordination

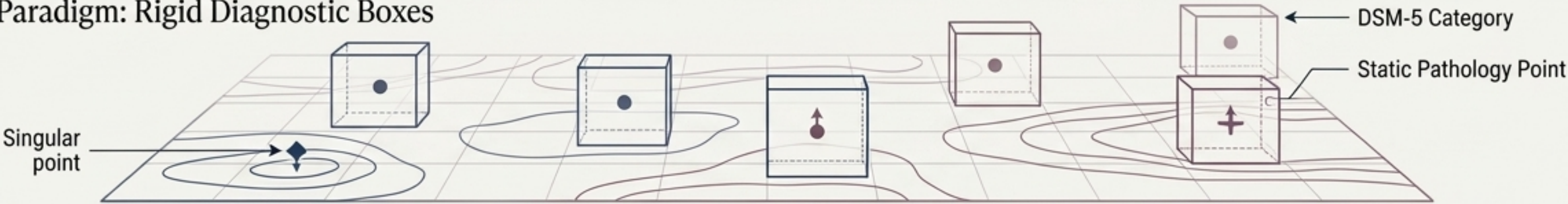
Without Central Control

Coherent global behavior emerges purely from local coupled field dynamics and spatial geometries.

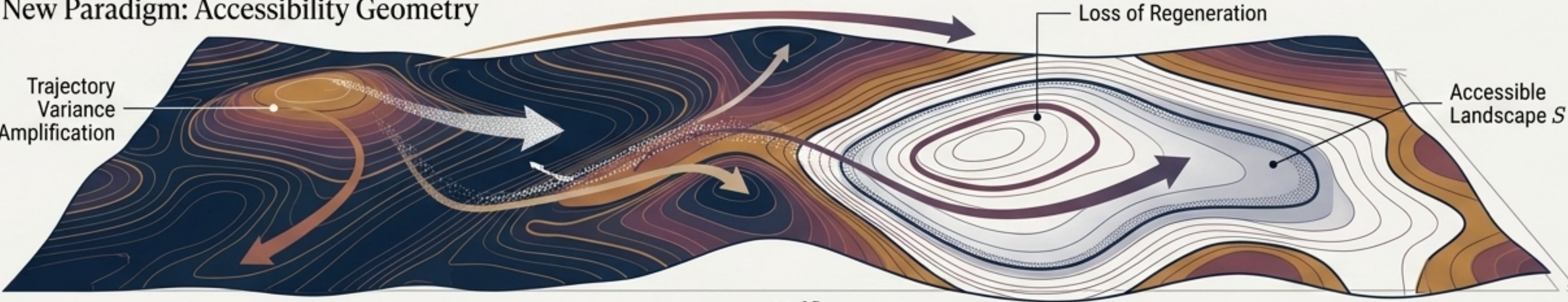
Paradigm Shift I: Psychiatry & Biomarkers

The RSVP framework demands a shift from cross-sectional configurations to longitudinal variance tracking.

Old Paradigm: Rigid Diagnostic Boxes



New Paradigm: Accessibility Geometry



$$S = \iint \frac{dS}{dt} dt$$

$$\nabla_t f(x, t)$$

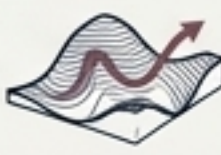
Current Assumption

Diagnoses assume point-estimates of static pathology.



RSVP Reality

Conditions are **geometric** deformations (trajectory variance amplification or loss of regeneration).



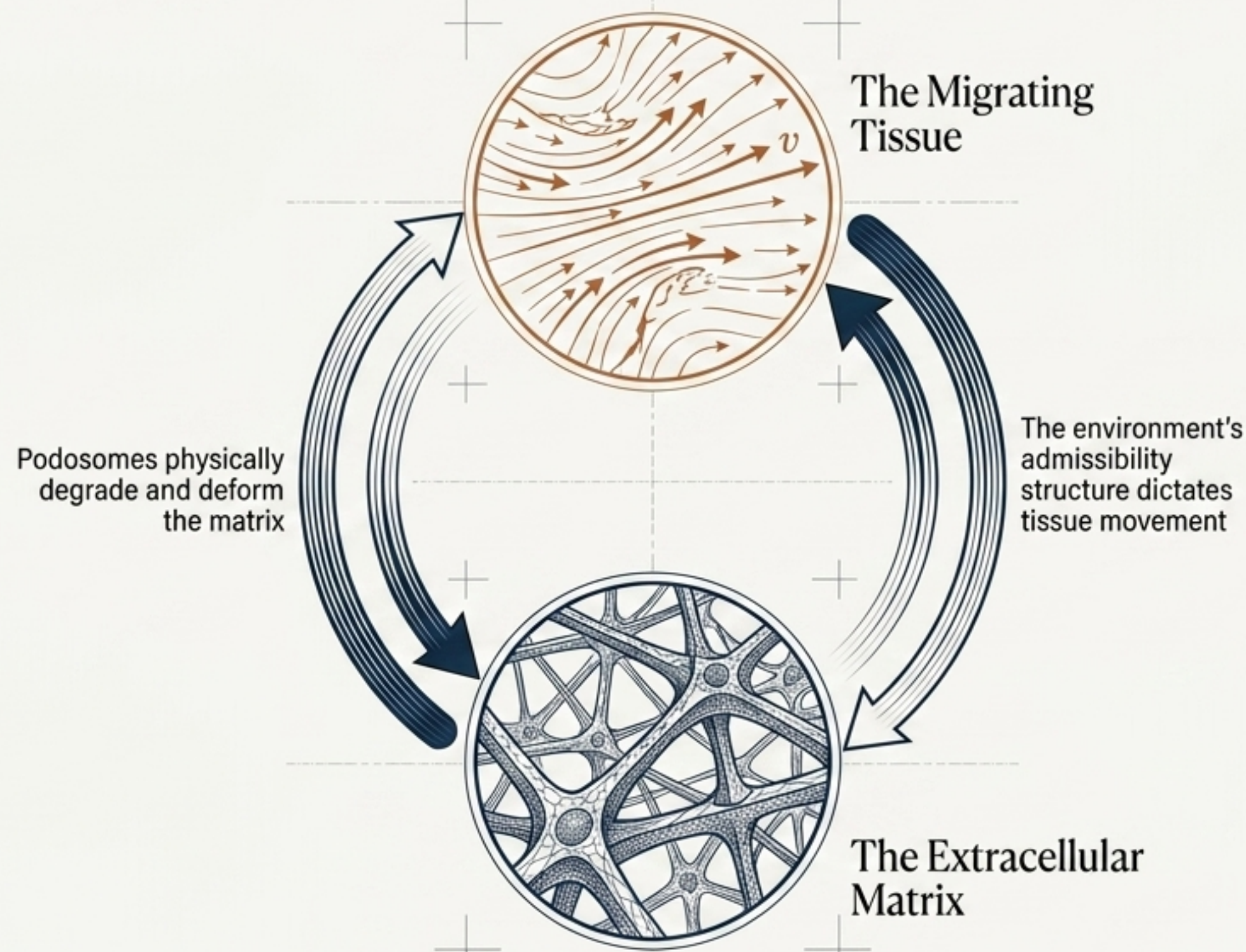
Future Biomarkers

Must estimate local dimensionality, **curvature**, and accessibility volume (S) longitudinally.

$$R_{ei}^t = \sum \frac{dS}{dt} \bigg|_t$$

Paradigm Shift II: Collective Intelligence

Tissues operate as decentralized computational systems, continuously rewriting the environmental constraints they inhabit.



Fundamental modeling limitation: The environment can no longer be treated as a fixed external input. The system's activity rewrites its own admissibility landscape.

Signature Predictions

Nontrivial, structurally motivated predictions that empirically discriminate RSVP from localist accounts.

1. Variance Transience	GAD PAD variance ($\sigma_{PAD}^2(t)$) will transiently widen during developmental milestones, not drift monotonically.
2. Multimodal Substructure	Symptom-matched cohorts will diverge into distinct consolidation-dominated vs. plasticity-dominated trajectories.
3. Topology over Amplitude	Wave-structured ERK stimulation will yield higher migration coherence than amplitude-matched static stimulation.
4. Wave-Order Hysteresis	Prior wave traversals permanently lower future ZO-1 redistribution thresholds ($\theta_n < \theta_0$).
5. Variance Precedes Mean	In any coupled-field perturbation, variance amplification systematically precedes mean-field shifts.

Biology as Dynamic Admissibility Geometry

The future lies in measuring distributed organizational dynamics that dictate what is possible.

Biological systems achieve macroscopic coherence through the propagation of redistributing dynamics across coupled fields, not through the aggregation of fixed object properties.

$$\frac{\partial^2}{\partial t}(\sigma_n(t)) = -\frac{1}{\partial v} \left(-\frac{\partial \sigma_{v,n}(t)}{\partial t} \right)$$
$$\frac{\partial^2}{\partial t}(\sigma_v \beta) = \alpha \frac{1}{\partial t} \left(-\frac{\partial \sigma_{v,n}(t)}{\partial t} \right) + \mu_v(\alpha \leq \theta)$$

$$\frac{\partial F}{\partial t} \cdot \frac{\partial F}{\partial t} = \frac{\sigma^2 \sigma_{PAD}(t)}{\partial_{PAD}}$$

Admissibility Gradient

$$\frac{\partial^2}{\partial t} = \frac{\partial^2}{\partial t} \frac{\partial^2}{\partial t} \frac{\partial^2}{\partial t}$$

Flow Potential

Variance Threshold

Variance Threshold